

# Quadratic Functions and Vertical Motion

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# The Physical Model

## Assumptions

Treat the ball as a point mass, neglect air resistance, and take gravitational acceleration  $g$  as constant. Choose height positive upward. Initial height and vertical velocity are  $s_0$  and  $v_0$ .

$$h(t) = s_0 + v_0 t - \frac{1}{2}gt^2, \quad v(t) = v_0 - gt.$$

This approximation describes motion close to the Earth's surface over a range in which variations of  $g$  are negligible.

# Quadratic Form and Maximum Height

## The Vertex

For  $y = ax^2 + bx + c$ , with  $a \neq 0$ , the vertex is at  $x = -b/(2a)$ . In vertical motion,

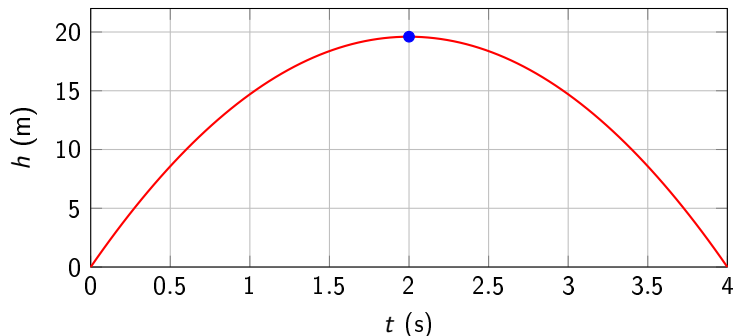
$$a = -g/2, \quad b = v_0, \quad c = s_0.$$

For an upward launch,  $v_0 > 0$ , the maximum occurs at

$$t_{\max} = v_0/g, \quad h_{\max} = s_0 + v_0^2/(2g).$$

At this time  $v = 0$ , and the negative quadratic coefficient makes the stationary point a maximum.

# Calculated Trajectory



The curve is calculated from  $h(t) = 19.6t - 4.9t^2$ . The modeled ball returns to ground level at  $t = 4$  s.

## Exercise

A ball is launched vertically from ground level with initial speed  $v_0 = 19.6$  m/s. Neglect air resistance and use  $g = 9.8 \text{ m/s}^2$ .

- 1 Determine the time at which it reaches maximum height.
- 2 Calculate the maximum height.
- 3 Calculate the maximum gravitational potential energy for mass  $m = 0.35$  kg, taking potential energy as zero at ground level.

## Solution

$$t_{\max} = \frac{19.6}{9.8} = 2 \text{ s.}$$

$$h_{\max} = 19.6(2) - 4.9(2)^2 = 19.6 \text{ m.}$$

$$E_{p,\max} = mgh_{\max} = 0.35(9.8)(19.6) \simeq 67.2 \text{ J.}$$

To two significant figures, the result is 67 J. The energy equals the initial kinetic energy,  $mv_0^2/2 = 67.228 \text{ J}$ . The quoted input precision does not justify retaining all those digits in the final physical result.

# Adding Horizontal Motion

For a projectile launched at angle  $\theta$ , with the same idealizations,

$$x(t) = v_0 \cos \theta t, \quad y(t) = s_0 + v_0 \sin \theta t - \frac{1}{2}gt^2.$$

## Range for Equal Launch and Landing Heights

If  $s_0 = 0$  and the projectile lands at  $y = 0$ , its range is

$$R = \frac{v_0^2 \sin(2\theta)}{g}.$$

At fixed speed, this is maximal at  $45^\circ$ . The conclusion need not hold for unequal heights or when air resistance is included.